

Reward Correctness vs. PPO Learnability in Walker-2D Hurdle Locomotion

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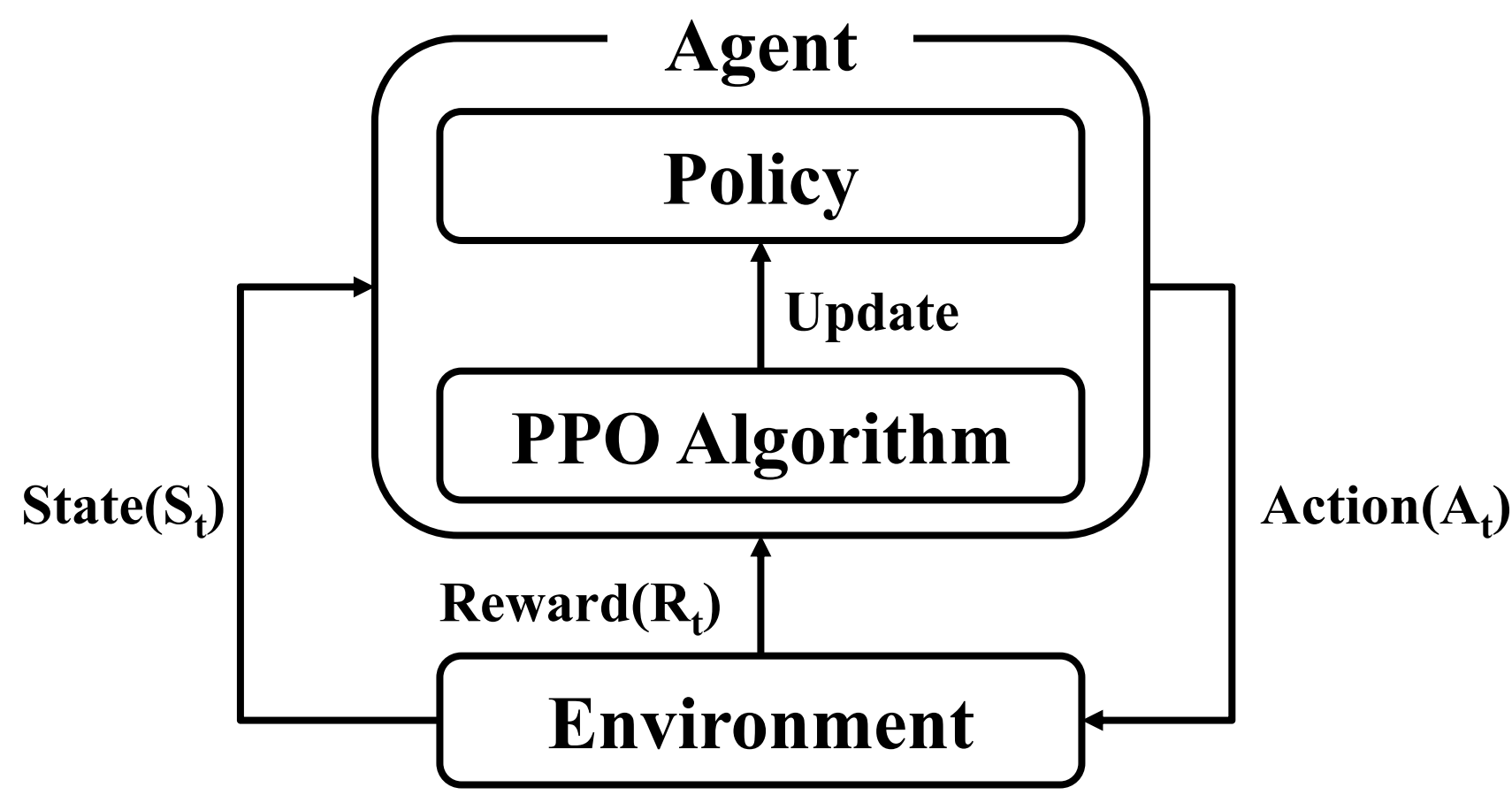
Background

Reinforcement learning

Trains an agent to learn a reward-maximizing policy through trial-and-error interaction with an environment.

Proximal Policy Optimization(PPO)

An actor-critic policy-gradient method that updates a stochastic policy from sampled trajectories while clipping large policy changes for stable learning.



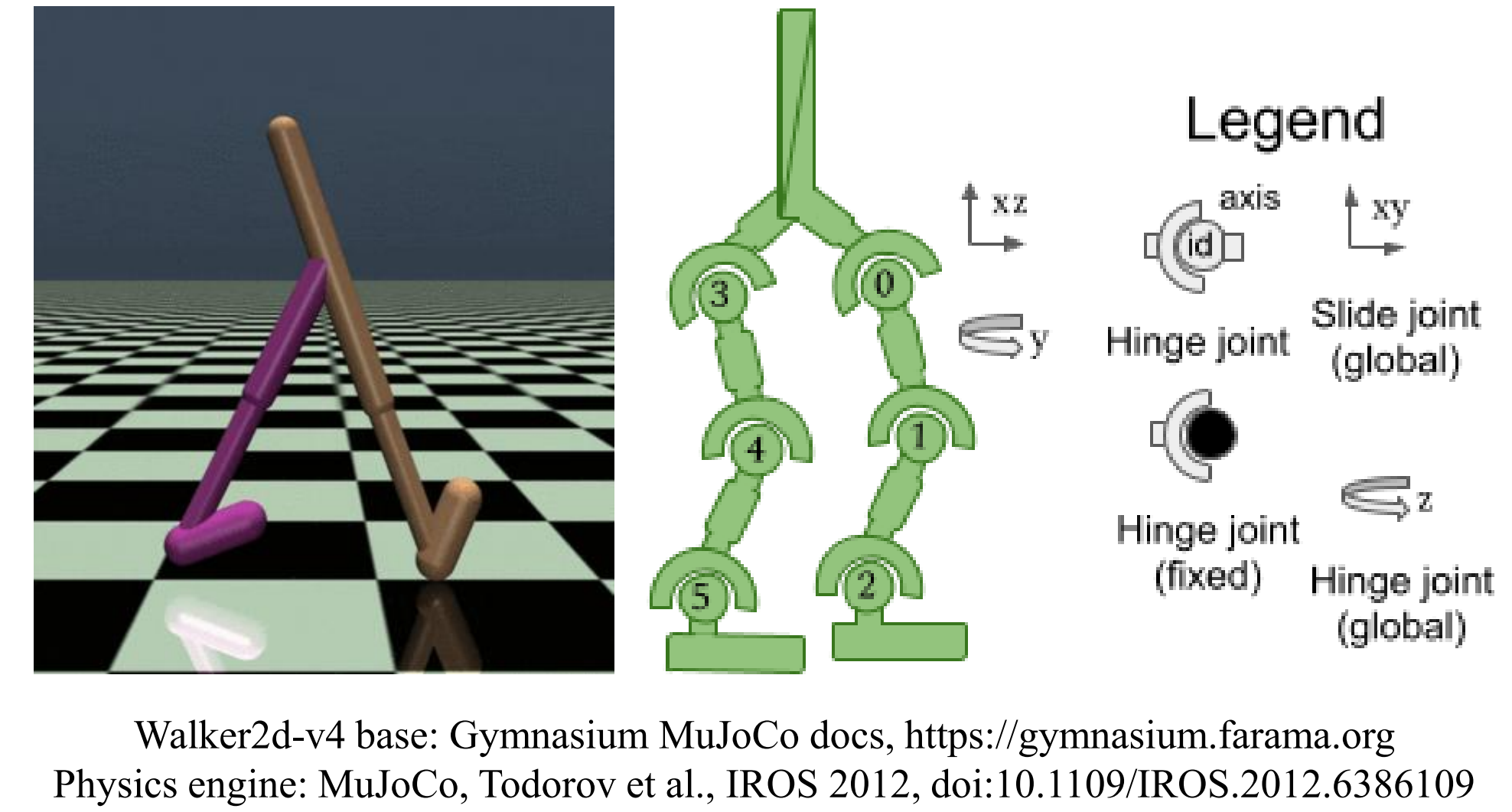
Gymnasium & MuJoCo

Gymnasium: standardized RL benchmark environments

MuJoCo: simulates rigid-body physics control tasks

Walker-2D

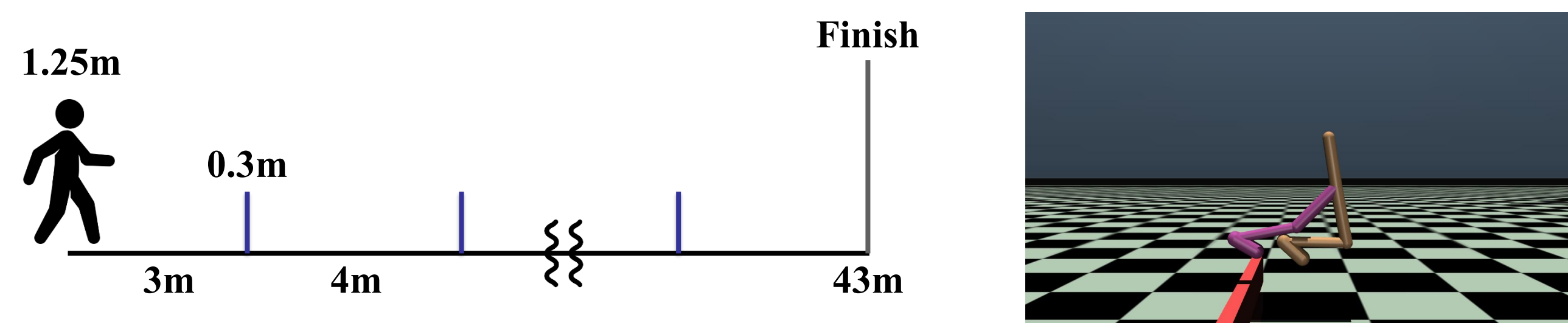
A 2D biped benchmark for forward locomotion without falling.



Methodology

Walker-2D-Hurdle Problem

- 10 free-body hurdles
- Episode ends by finish, fall, disqualification, or timeout
- Aims for **Qualified-and-Fast finish**



Olympic-Style Legal Pass Rule

- A hurdle is counted only when both right and left foot-leg pairs cross the hurdle line above the original 0.30 m hurdle top.
- Under-hurdle crossing or bypassing is treated as failure.

Reward Design

$$R_t = R_{forward} + R_{healthy} + R_{control} + R_{legal-pass} + R_{clearance} + R_{finish}$$

Component	Purpose	Formula
Forward velocity	Encourage forward locomotion	V_x
Healthy reward	Maintain upright stable gait	$t * w_h$
Control cost	Minimize useless motion	$-0.001 * [\#controls]$
Legal pass reward	Reward valid hurdle clearance	$30 * [\#legal\ pass]$
Clearance shaping	Guide feet/legs above hurdle	$\exp(-2g^2) *$
Finish bonus	Reward fast completion	$\max(0, 1000 - 0.5t) **$

* $g = \max(0, h_{hurdle} + 0.15 - z_{pair})$
 z_{pair} : height of the best-lifted foot-leg pair, measured by the lower of its foot and leg height
**Timeout set to 2500 episodes for DNF

Training

Used provided Stable-Baselines3 PPO training pipeline with vectorized normalized environments.

Evaluation

Deterministic evaluation with task-specific metrics:

- Finish rate
- Mean legal hurdles passed
- Disqualified / Fall / Timeout rate
- Finish time among successful episodes

Prioritized legal task completion before reward or speed.

Results

Final Policy Performance

Training : 9M, Evaluation : 500 episodes

Finish rate	DQ rate	Timeout rate	Mean legal hurdles	Mean finish time
76,8%	0.2%	0.8%	8.81/10	12.13s

The final policy reliably completed the legal 10-hurdle task, with low disqualification and timeout rates.

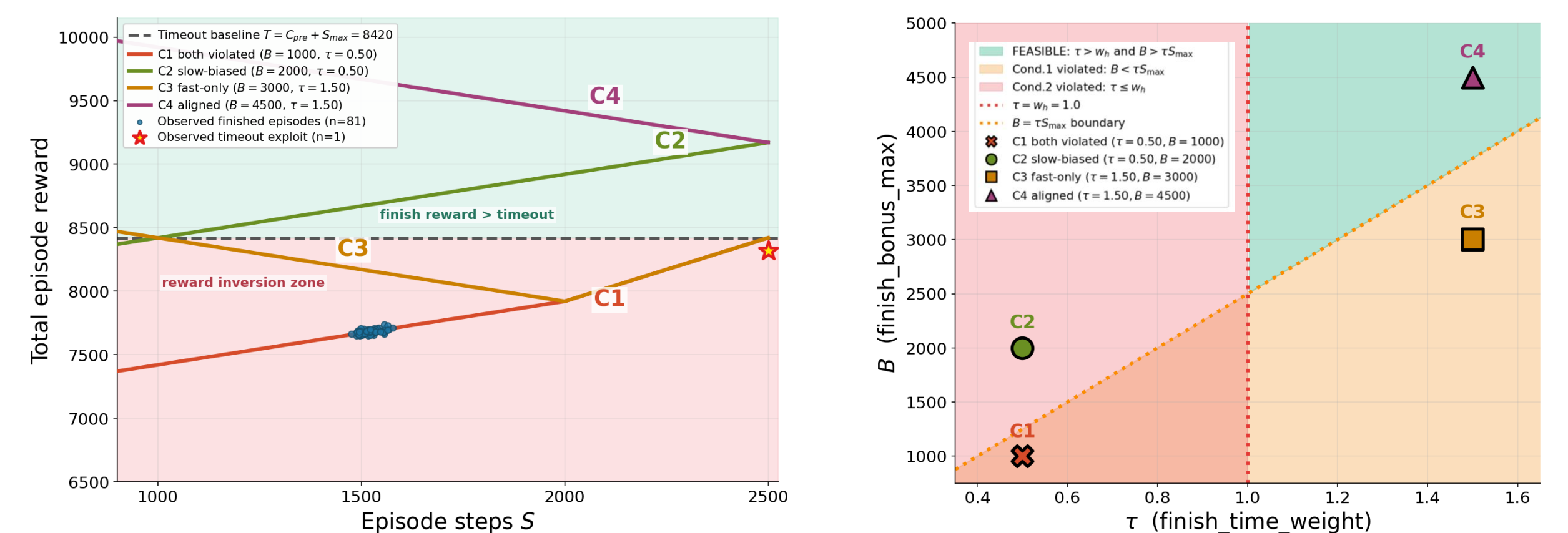
Mathematical Reward Correctness

$$f(S) = C_{pre} + R_{healthy} + B - \tau S$$

C_{pre} : Reward independent to step, B: Instant finish reward, τ : Time decay rate

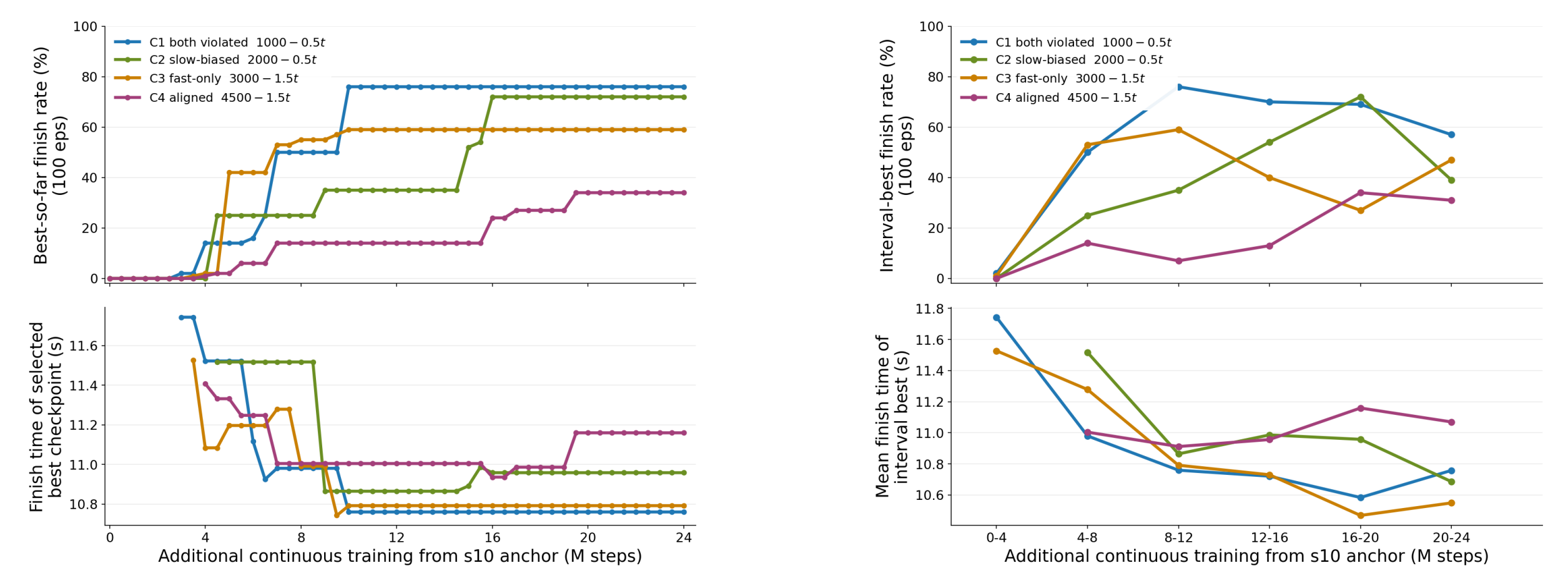
Condition 1. Finish should beat timeout : $B > \tau S_{timeout}$

Condition 2. Faster finish should be preferred: $\frac{dB}{dS} = w_h - \tau < 0$



Reward Correctness vs. PPO Learnability

100 episodes evaluation after continuous training from before-finished 5M trained seed(s10 anchor).



Mathematically aligned C4 was not the easiest objective for PPO

PPO performance depended strongly on optimization accessibility

Conclusion

- Final policy achieved 76.8% finish rate over 500 deterministic evaluations.
- A mathematically aligned reward defines the right target, but PPO may learn better from a flawed reward landscape that is easier to optimize
- Reward correctness is not the same as PPO learnability.